



Beyond Steps: Unlocking Multimodal Health Discovery with the World's Largest Dataset

February 13, 2026

Josh Denny, M.D., M.S.
Chief Executive Officer

Factors of Risk in the Development of Coronary Heart Disease— Six-Year Follow-up Experience

The Framingham Study

WILLIAM B. KANNEL, M.D., THOMAS R. DAWBER, M.D., F.A.C.P.,
ABRAHAM KAGAN, M.D., F.A.C.P., NICHOLAS REVOTSKIE, M.D.,
AND JOSEPH STOKES, III, M.D.
Framingham, Massachusetts



Framingham Heart Study

Enrolled 5,209 men and women in 1948

Some early Framingham discoveries:

The vision: have the same impact
with wearables, molecular data,
precision medicine, and AI in the next
<<60 years.

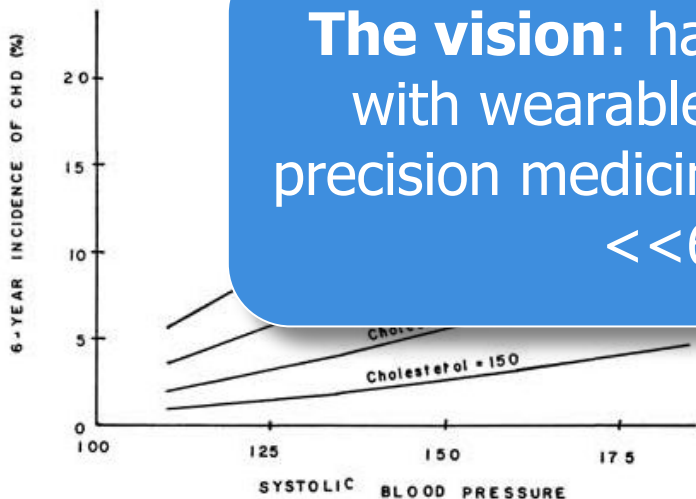
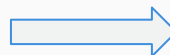


FIGURE 2. Six-year incidence of coronary heart disease according to level of systolic blood pressure at specified serum cholesterol levels (men 45 to 62 years). For explanation, see legends for Figure 1.

***Interventions have resulted in a >60%
reduction in cardiovascular mortality***

A Quick Reminder from an Early “Wearable”: the ECG

In the 1980s, abundant ECGs and research taught us about the risk of ventricular ectopy



Myocardial infarction
Sudden cardiac death
Arrhythmias

If ectopy is bad and antiarrhythmics effectively eliminate ectopy, shouldn't everyone with ectopy be on antiarrhythmics?

The New England Journal of Medicine

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Volume 324

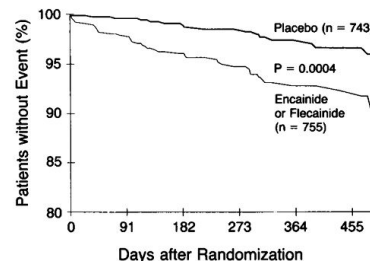
MARCH 21, 1991

Number 12

MORTALITY AND MORBIDITY IN PATIENTS RECEIVING ENCAINIDE, FLECAINIDE, OR PLACEBO

The Cardiac Arrhythmia Suppression Trial

DEBRA S. ECHT, M.D., PHILIP R. LIEBSON, M.D., L. BRENT MITCHELL, M.D., ROBERT W. PETERS, M.D.,
DULCE OBIAS-MANNO, R.N., ALLAN H. BARKER, M.D., DANIEL ARENSBERG, M.D., ANDREA BAKER, R.N.,
LAWRENCE FRIEDMAN, M.D., H. LEON GREENE, M.D., MELISSA L. HUTHER,
DAVID W. RICHARDSON, M.D., AND THE CAST INVESTIGATORS*

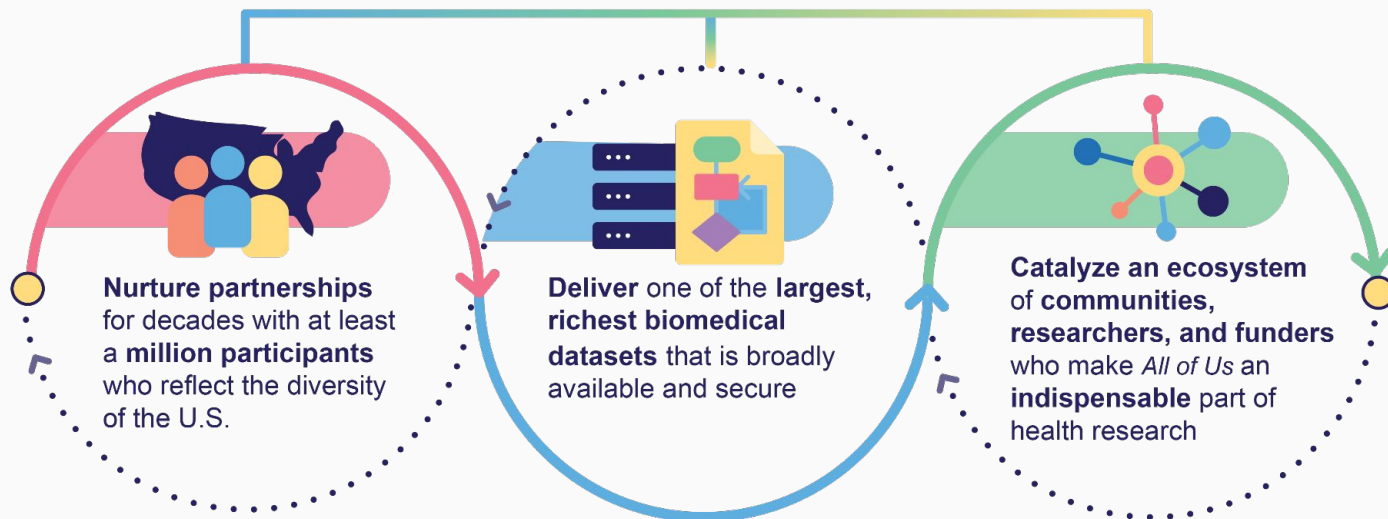


Placebo	743	632	516	412	292	201
Active drug	755	631	507	392	286	198

Figure 1. Actuarial Probabilities of Freedom from Death or Cardiac Arrest Due to Arrhythmia in 1498 Patients Receiving Encainide or Flecainide or Corresponding Placebo.

The *All of Us* Research Program Mission

Accelerate health research and medical breakthroughs,
enabling individualized prevention, treatment, and care for all of us



Made possible by a team that maintains a culture built around the program's core values

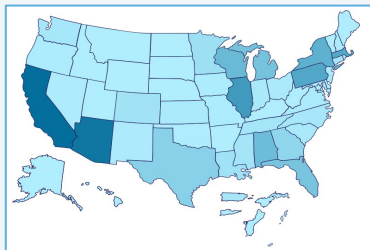
Overview of the *All of Us* Research Program

What Goes in...



Data from **874,000+** Participants from all 50 states including:

14 Million+ Biosample aliquots
475,000+ EHRs
2.9 Million+ Survey Responses



Gets Combined & Curated...



Data Refinements



Cloud-based Platform

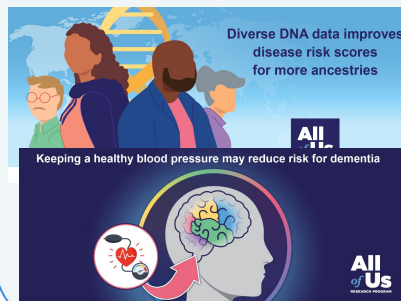
For harmonization,
participant privacy, and
quality

Leading to New Research...

From **21,350+**
Registered
Researchers
from **1,300+**
Organizations



1,180+
Publications
using *All of Us*
Data, such as:



And Valuable Health Information Returned to Participants

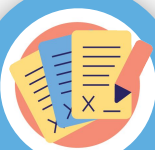


733,000+ Individual DNA
Results Returned to **270,000+**
Participants, including:

5,490+ with potentially
life-changing genetic
variants in 59 genes

145,000+ that could
impact how their body
processes 50+ medicines

Data Currently in the *All of Us* Researcher Workbench



Participant Surveys

2,900,000+ Surveys Completed



Electronic Health Records

393,000+ EHRs



Physical Measurements

509,000+ Physical Measures



Biospecimens

14,000,000+ Aliquots (DNA, RNA, serum, plasma, cfDNA, urine)



Assays

97,000+ Structural Variants



Omics

447,000+ Genotyping Arrays
414,000+ Whole Genome Sequences
2,700+ Long-Read Sequences



Mobile/Wearable Tech

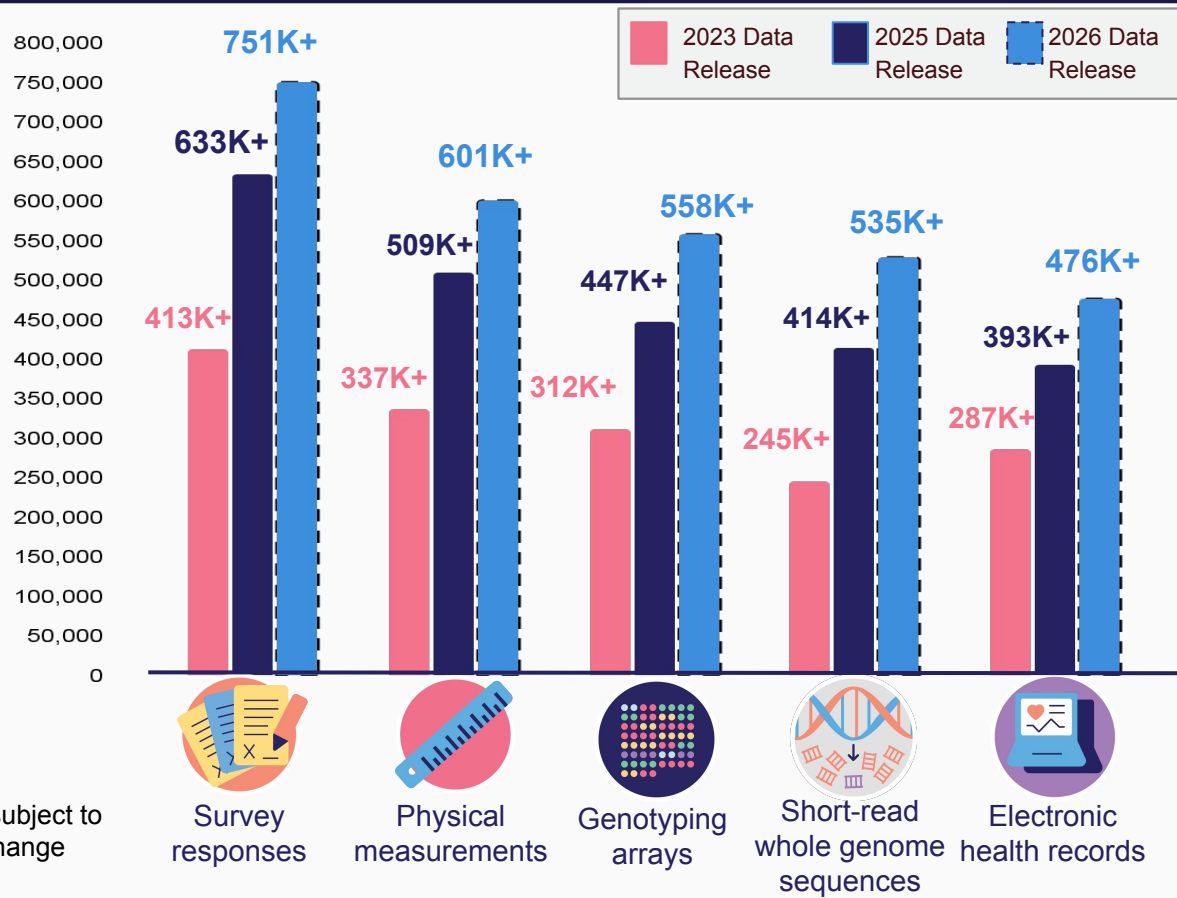
59,000+ Fitbit Records

The whole genome sequence dataset includes variation at more than **1.4 billion** locations, which is more than **one-third** of the entire human genome.



The *All of Us* Researcher Workbench contains the **world's largest research-accessible dataset of wearable data**, uniquely linked with clinical, genomic, and survey information

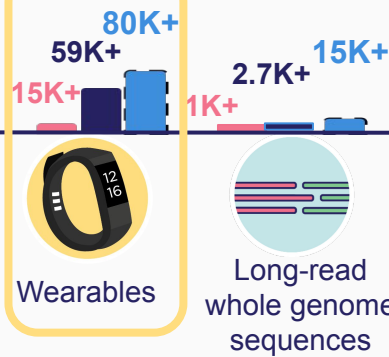
Refreshed Data for Researchers Every 1–2 Years



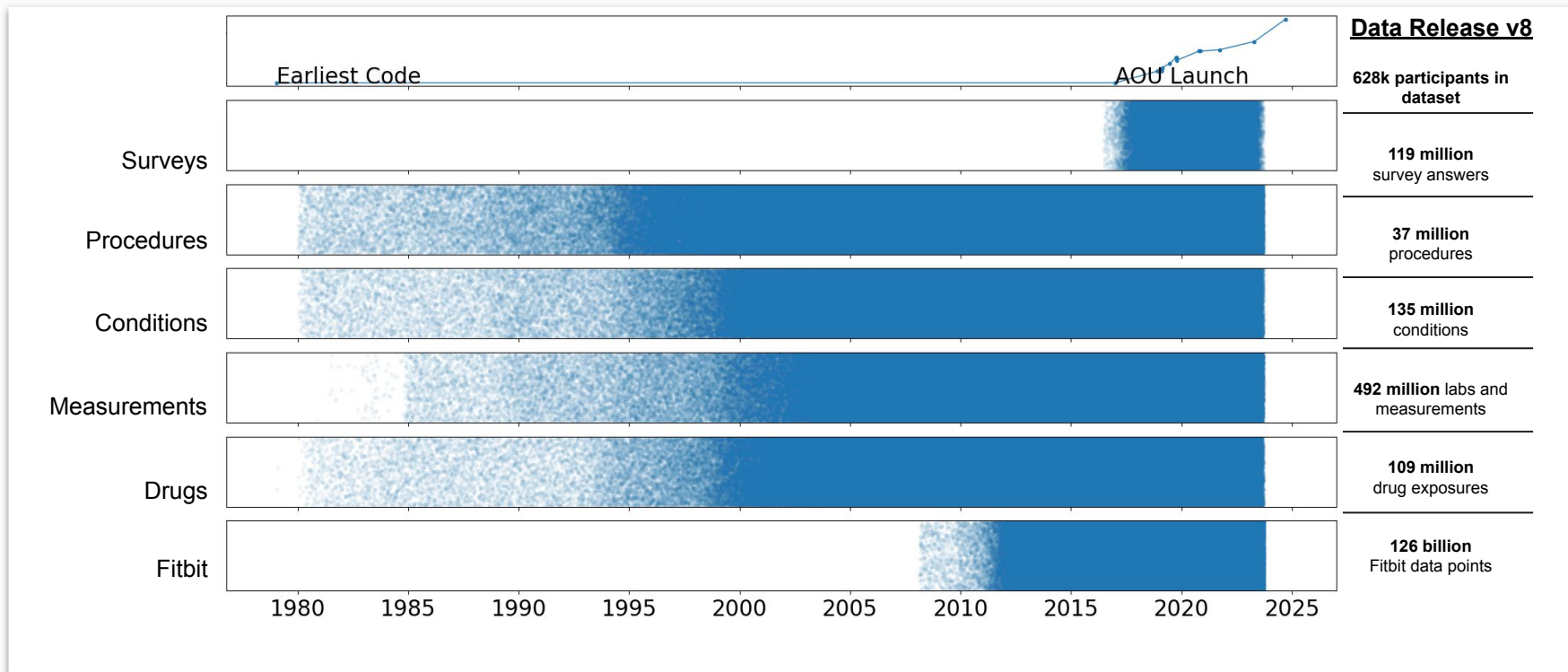
New Data Types in Future Releases

- Proteomics (10k+)
- Transcriptome/RNAseq (10k+)
- Metabolomics (5.5k+)
- Clinical Notes Pilot (48k)
- Olink Pilot (19k+)
- Exposomics pilots (6k+)

Wearables data from 80,000+ participants



Phenotypic Data on the Researcher Workbench is Diverse and Longitudinal



All of Us Can Be Used to Study a Wide Range of Diseases

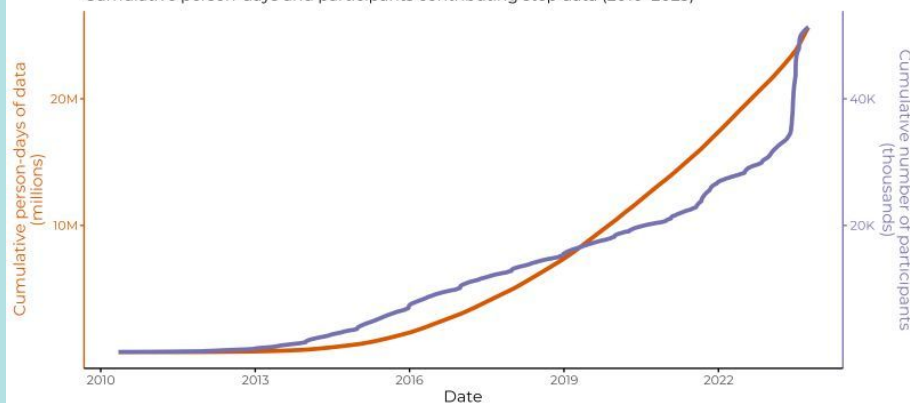
Conditions	Domain	Participants
Heart Disease	Heart	119,000
Obesity	Endocrine	111,560
Type 1 Diabetes	Endocrine	8,260
Type 2 Diabetes	Endocrine	74,920
Any cancer	Cancer	65,980
Asthma	Pulmonary	65,180
Chronic Obstructive Pulmonary Disorder	Pulmonary	32,020
Epilepsy	Neuro	11,680
Stroke	Neuro	1,460
Rheumatoid Arthritis	Autoimmune	11,380
Osteoarthritis	Autoimmune	132,800

Conditions	Domain	Participants
Depressive Disorder	Mental Health	112,440
Bipolar Disorder	Mental Health	20,540
Dementia	Mental Health	8,160
Human Immunodeficiency Virus	Infectious Disease	6,400
Alcoholism	Abuse	25,680
Substance abuse	Medication	57,920
Age Related Macular Degeneration	Eye	7,980
Hearing loss	Hearing	52,120
Falls	Aging/Nursing	9,180

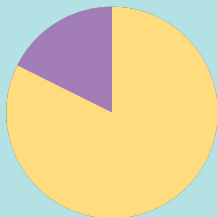
Wearables Data Growth

Fitbit data are in-depth and longitudinal...

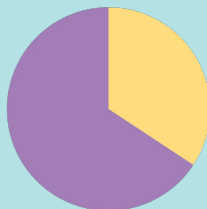
Every Step Counts: Fitbit Participation in the *All of Us* Research Program
Cumulative person-days and participants contributing step data (2010–2023)



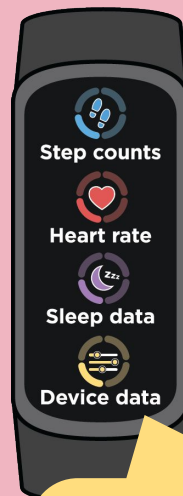
>82%
Provide all
types of Fitbit
data



>34%
Provide >3
years
worth of
Fitbit data



...And can be linked with other health data



**Survey
Responses**



**Physical
Measurements**

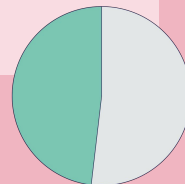


**Genomic
Data**



EHRs

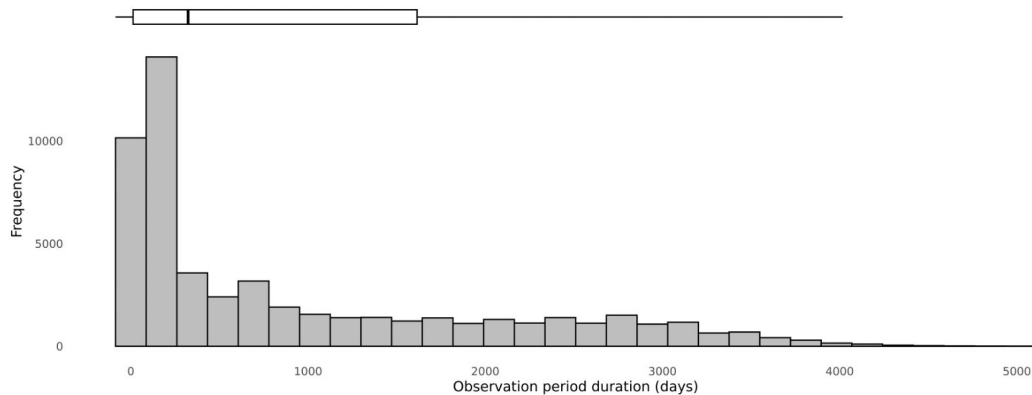
- FitBit Data includes:**
- Activity summaries
 - Heart rate (minute-level and summary data)
 - Steps intraday
 - Sleep level
 - Time spent in each sleep level



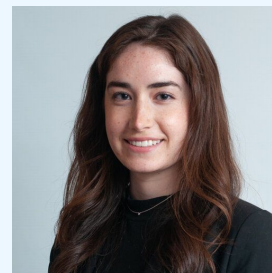
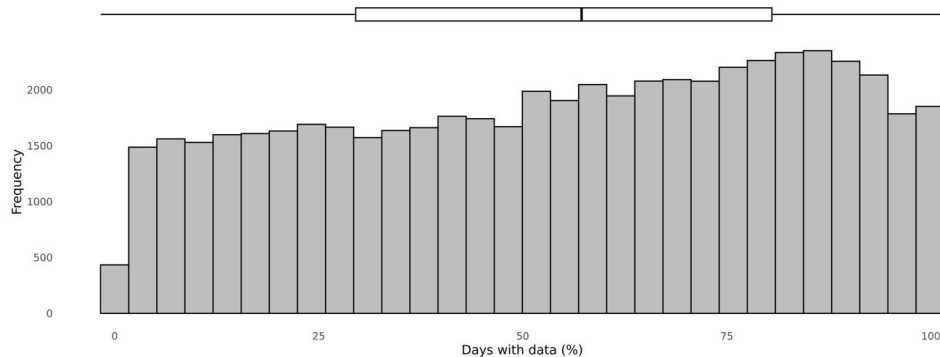
>48%
Provide Fitbit
+ all four data
types

Fitbit Data—Step Count

Duration of Fitbit Data



Percentage of Days with Data



Courtesy of Evelynne Fulda, **M.Sc.**
NIH/Oxford DPhil Student

Fulda et al. (2026) *MedRxiv*,
<https://doi.org/10.64898/2026.01.29.26344899>

All of Us Research Hub: Public Data Browser



Summary statistics of:

- EHR Data (Conditions, Drug Exposures, Lab & Measurements, Procedures)
- Genomic Variants
- Survey Questions
- Physical Measurements
- **Open Access (no login required)**

Data Browser



Search Across Data Types ⓘ



Cancer



Data includes 633,540 participants as of 10/1/2023.

FAQ

Electronic Health Records (EHR)

Conditions ⓘ

215

matching

354,400 p

View Con

Labs & Measurements ⓘ

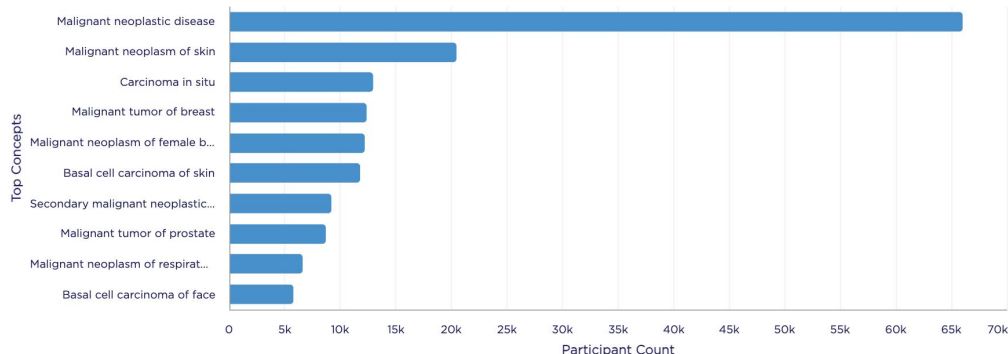
170

Procedures ⓘ

50

DATA DISCLAIMER

Top 10 Conditions by Descending Participant Counts ▾



Surveys

Personal
History

23

matching s

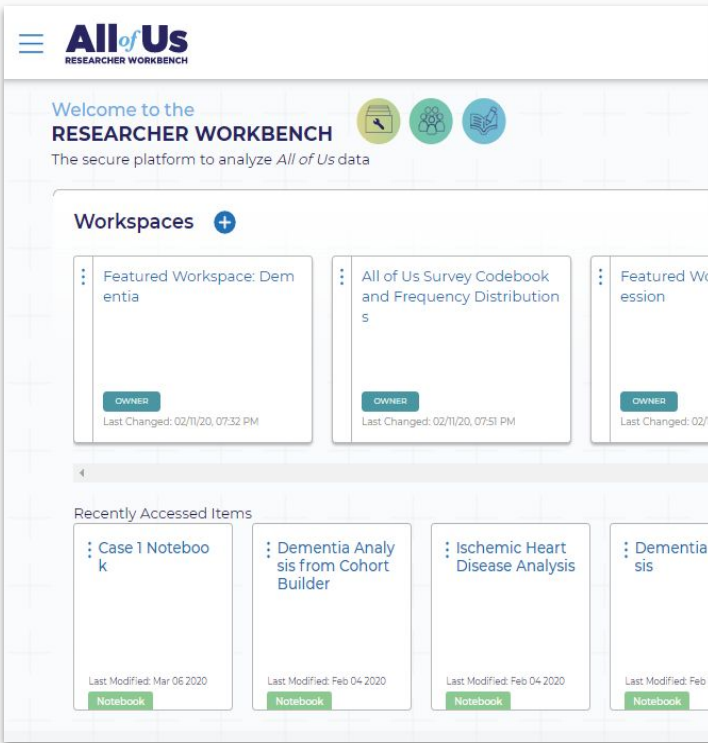
298,100 p

All of Us Researcher Workbench: Access to Row-Level Data for Analysis



Researcher Workbench

- Cloud-based platform
- Graphical tools for creating cohorts and datasets
- Open to researchers at academic, not-for-profit, and health care institutions
- **Passport access model** with institutional agreements for all researchers
- **No separate IRB approval needed**



Explore the data using:





ResearchAllofUs.org/Data-Tools/Workbench/

Rapid Access to Data & Tools: Test Your Hypothesis This Week!

(data as of February 2, 2026)



21,350+ Registered Researchers



1,300+ Organizations

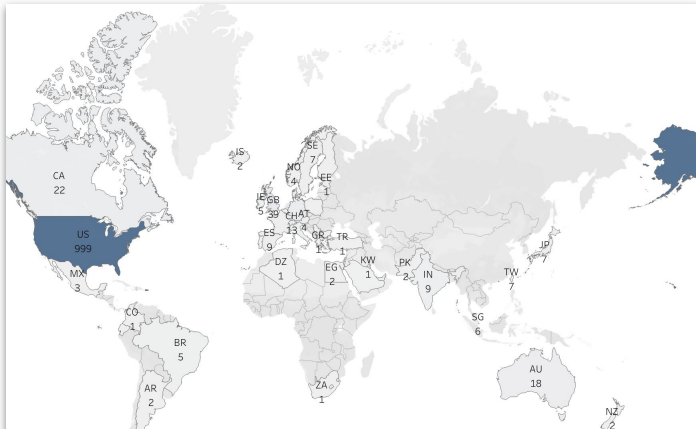


22,440+ Workspaces Created



1,180+ Publications
using *All of Us* data

All of Us is open to national and international researchers at academic, not-for-profit, commercial, and health care institutions.



**1,120+ International researchers,
370+ from the UK**

40+ UK Organizations including:

GENOMICS



**Imperial College
London**



precision**life**



wellcome
sanger
institute



Sequoia
GENETICS

All of Us Data Leading to New, Impactful Health Discoveries

Fitbit and wearables data are being used to make new discoveries in population health—from hospital readmission, to chronic disease risk, to guiding physical activity recommendations

A collage of scientific articles related to wearable devices and health data. The articles include:

- Molecular Psychiatry**: Integrating genome-wide information and wearable device data to explore the link of anxiety and antidepressants with pulse rate variability. Published: 18 November 2024.
- Appl Clin Inform**: Predicting Postoperative Pain and Opioid Use with Machine Learning Applied to Longitudinal Electronic Health Record and Wearable Data.
- PNAS**: Sleep duration and timing are associated with next-day physical activity: Insights from two large-scale wearable sensor studies.
- JCEM**: Predicting incident type 2 diabetes using wearable activity and polygenic risk: A survival-modeling study in All of Us.
- JMIR mHealth and uHealth**: Predicting Hospital Readmission among Patients with Sepsis Using Clinical and Wearable Data.
- JMIR mHealth and uHealth**: Risk for Poor Post-Operative Quality of Life Among Wearable Use Subgroups in an All of Us Research Cohort.
- WILEY**: Physical Activity, Sleep, and Quality of Life in Pulmonary Arterial Hypertension: Novel Insights From Wearable Devices.
- JMIR mHealth and uHealth**: Harnessing Consumer Wearable Digital Biomarkers for Individualized Recognition of Postpartum Depression Using the All of Us Research Program Data Set: Cross-Sectional Study.

1,180+ scientific studies published, with over 500 in 2025

NEW Publication Directory

Publication Directory

Explore publications based on Researcher Workbench projects, program analyses, and more

Q Enter keyword

SEARCH

FILTERS

Type

- ☒ All
- ☐ Original Study
- ☐ Methods Paper
- ☐ Secondary Study
- ☐ Commentary / Review

Date Range

--Year-- --Month--
- THROUGH -
--Year-- --Month--

1,187 Publications

Health Literacy and Awareness of Family Health History in the All of Us Research Program

Jemar R Bather, Melody S Goodman, Stephanie H Cook, Kimberly A Kaphingst

Public Health Genomics
Published: January 2026
PubMed: 41553968

Show AI Summary

Show Abstract

Dysphonia Is Associated With Anxiety Research Program

Hannah N W Weinstein, Michael V Golub, Hayley L Born



Anyone can ask & answer important scientific questions – *All of Us* democratizes biomedical research

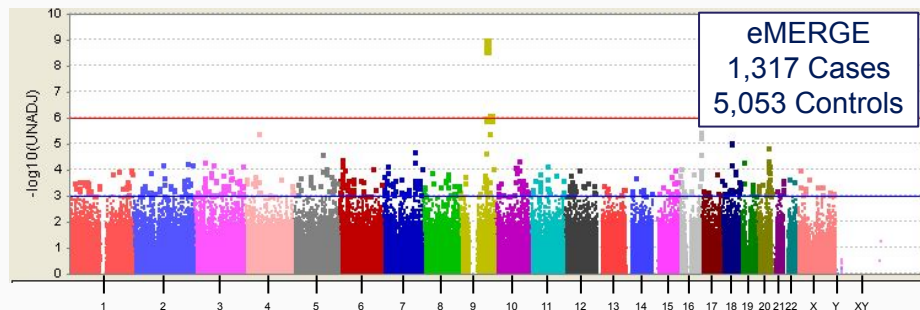


Anav Babbar
Former postbacc &
current med student

One Example: The Genetic Basis of Primary Hypothyroidism

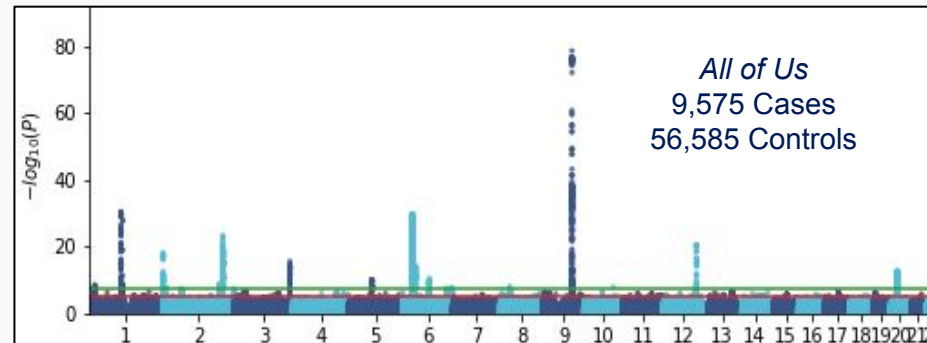
2011: NHGRI eMERGE Network Study

- Discovered one gene associated with hypothyroidism in a study group composed only of people with European ancestry
- Time and expense: 40 people
2.5 years
5 sites, each with \$1M/yr award

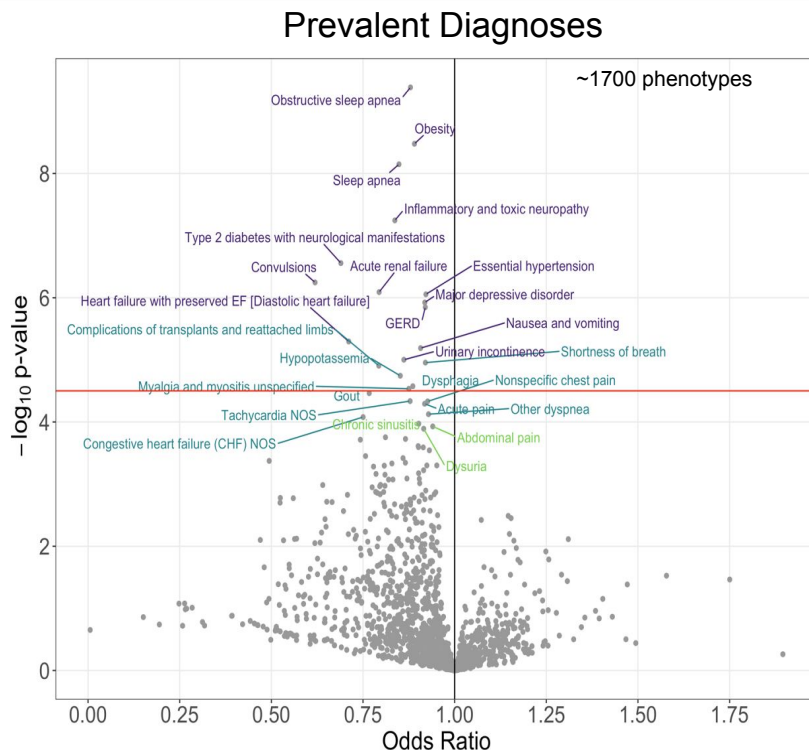


2023: *All of Us* Researcher Workbench

- Found 10+ unique genes associated with hypothyroidism using *All of Us* population
- Time and expense: 1 postbacc student
6 weeks
~\$40 in computing costs



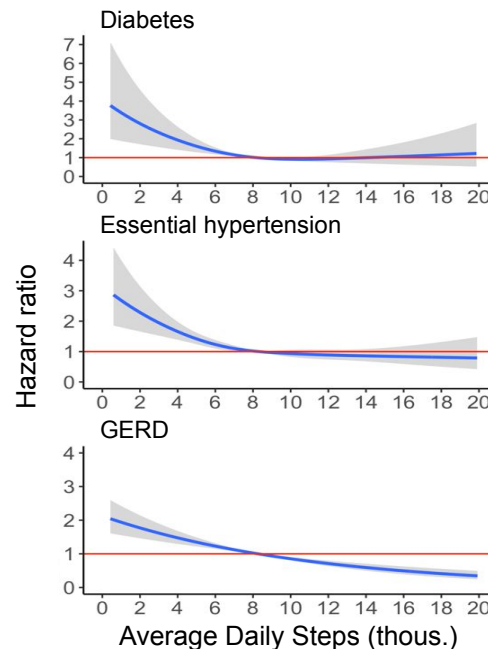
All of Us Discoveries: Physical Activity and Human Disease



- Analysis uses average of daily steps.
- Effect size per 1000 steps, adjusted for age, sex, race

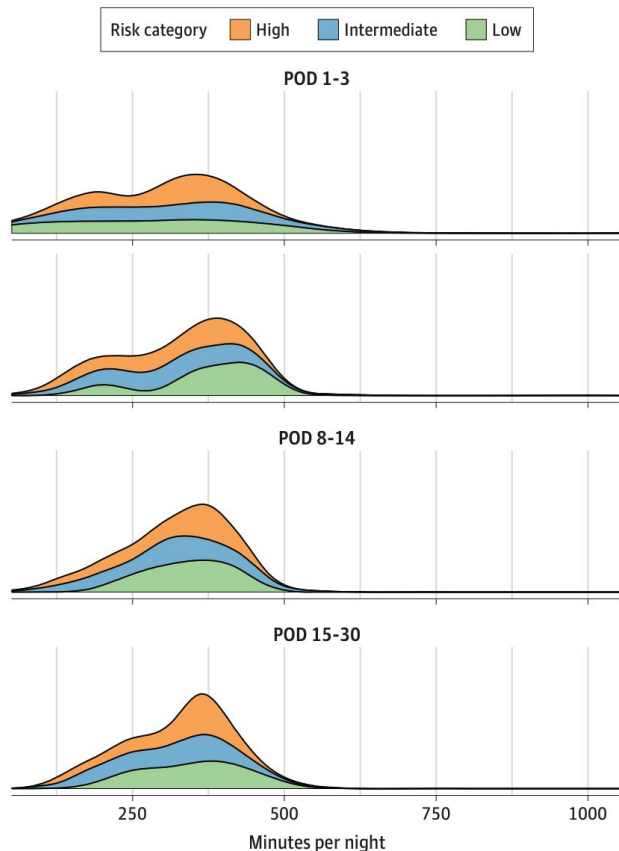
Master et al. (2022) *Nat Med*, 10.1038/s41591-022-02012-w.

Risk of incident disease after 6 months



Adjusted for age, sex, race, CAD, cancer, BMI, systolic blood pressure, education level, smoking, alcohol use, and *monthly daily steps (time varying)*

All of Us Discoveries: Wearables Inform Postoperative Risk



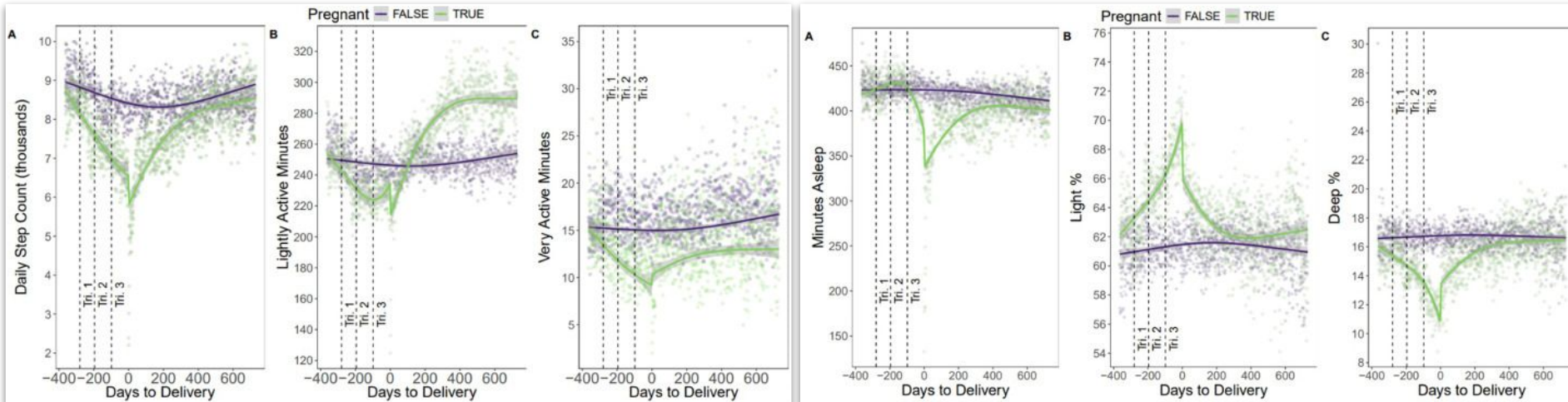
- Researchers analyzed **634 surgical procedures from 512 All of Us participants** who had donated Fitbit data. ~19% of procedures were classified as low risk, ~33% as intermediate risk, and ~48% as high risk.
- **High-risk procedures were associated with reductions in REM and deep sleep through postoperative day 7.**
- Intermediate-risk procedures were associated with milder disruptions, primarily limited to days 0 through 3.
- In contrast, low-risk procedures were not associated with significant changes in any sleep metric at any postoperative epoch.
- **Opioid exposure was associated with increased light sleep but suppressed deep sleep.**

All of Us Discoveries: Wearables and Pregnancy

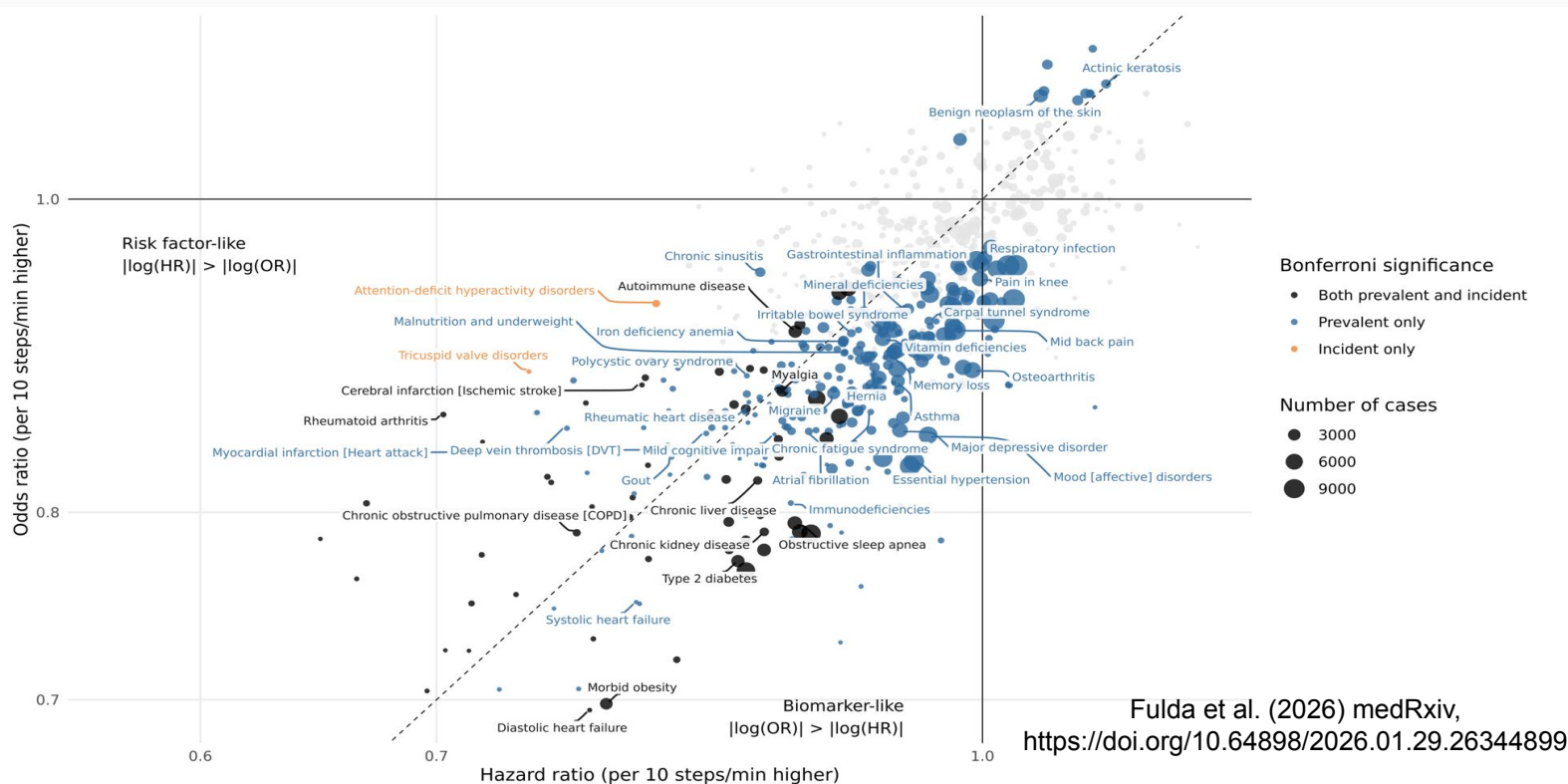
- Few large, diverse datasets have tracked changes in activity and sleep during and after pregnancy
- Recovery to prepregnant steps/day took 591 days. There was similar interruption in sleep duration, beginning in the first trimester, with less deep sleep in the first ~year after delivery

Activity

Sleep



Comparison of incident and prevalent associations with peak 30-min cadence



Evelynne Fulda, M.Sc.
 NIH/Oxford DPhil Student

Because of *All of Us*, There is a New Clinical Test

- **Polygenic risk scores** can help people understand their risk of disease by summing up the effects of many genetic variants.
- **Data from more than 236,000 *All of Us* participants** have helped researchers learn more about the link between differences in DNA and disease.
- **That data has led to a new clinical test for 8 cardiovascular conditions**

Learn
more



RESULTS SUMMARY

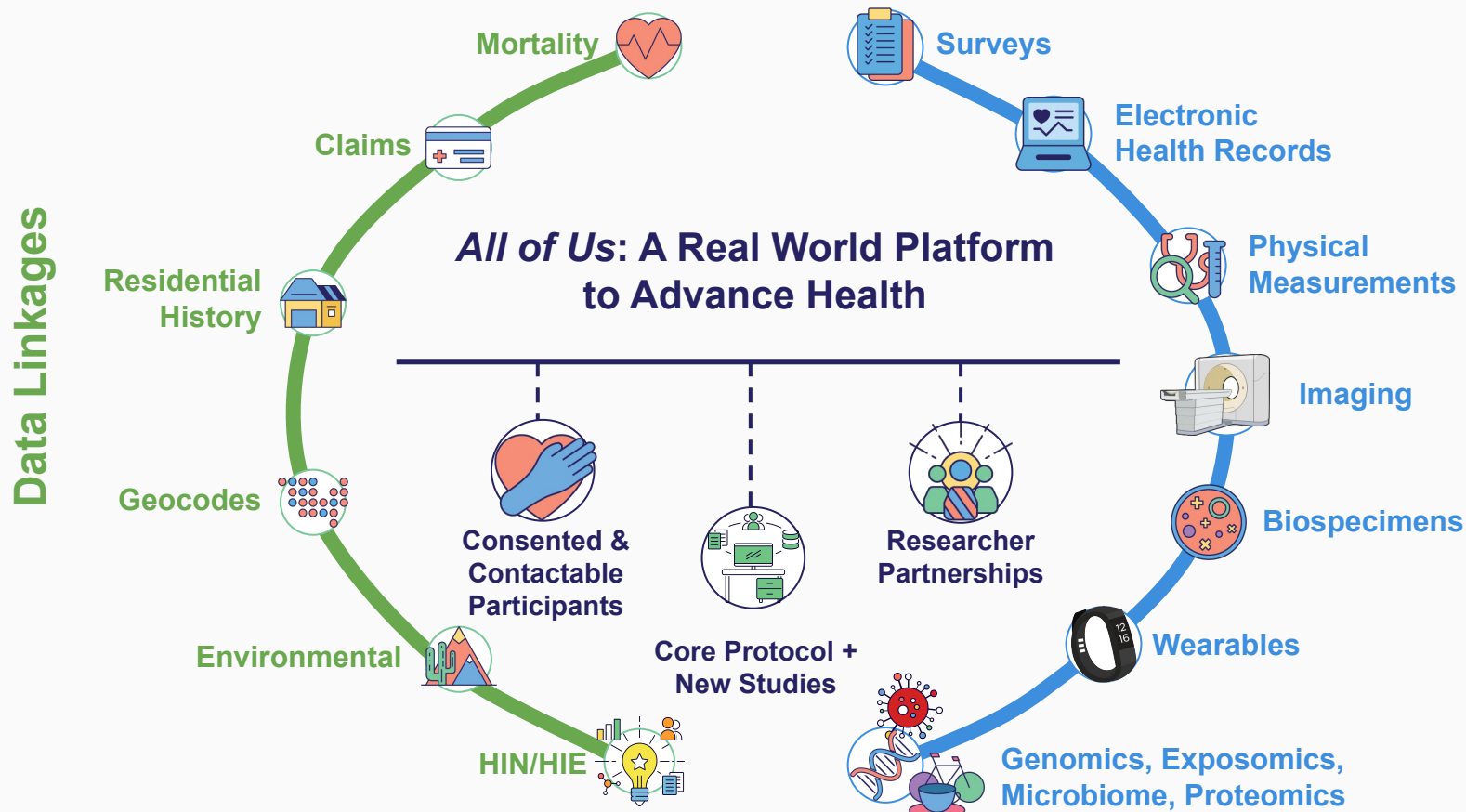
This individual was not found to be at high polygenic risk for the conditions assessed.

Polygenic risk describes the chance of developing certain health conditions based on a large number of genetic variants across the genome. This analysis does **not** examine rare variants in genes associated with monogenic forms of these conditions. Additional testing may be appropriate if the individual has clinical features or a family history of monogenic conditions. These results should be interpreted in the context of this individual's personal medical history and family history. For assistance in locating genetic counseling services or disease specialists and for questions about this report, please contact LMM@mgb.org.

POLYGENIC RISK SUMMARY

Atrial Fibrillation	Below Average	Average	Above Average	High
Coronary Artery Disease	Below Average	Average	Above Average	High
Diabetes Mellitus, Type 2	Below Average	Average	Above Average	High
Elevated lipoprotein(a)	Below Average	Average	Above Average	High
Hypercholesterolemia	Below Average	Average	Above Average	High
Hypertension	Below Average	Average	Above Average	High
Thoracic Aortic Aneurysm	Below Average	Average	High	
Venous Thromboembolism	Below Average	Average	High	

Data Linkages to Expand and Deepen Data



A Platform to Launch New Studies: Partnered Research Studies

Partnered Research Studies

- Participants may be **invited** to participate in another study, contribute new data, or generate new data from existing samples
- Studies may be relevant to everyone or a **smaller group of participants**
- Studies go through a **careful approval process**, including ethical and participant review



Learn More!

Ongoing and Future Studies



**Nutrition
for Precision
Health**

Studying how the body responds to changes in daily nutrition.



Studying environmental risk factors for type 2 diabetes.



National
Eye
Institute

Studying how retinal images uncover early signs of disease.



Studying how gene variations lead to specific outcomes.

Enabling Access to *All of Us* Biospecimens to Advance Science



Anticipated NOFO April 2026

Accelerating Discovery through Partnered Research with *All of Us* to Analyze Participant Biospecimens



Home Learn Grants ▾ Search Grants Applicants ▾ Grantors ▾ System-To-System ▾ Forms ▾ Connect ▾ Support ▾

VIEW GRANT OPPORTUNITY FORECAST

PAR-26-127

Accelerating Discovery through Partnered Research with All of Us to Analyze Participant Biospecimens

Department of Health and Human Services

National Institutes of Health

FORECAST

VERSION HISTORY

RELATED DOCUMENTS

PACKAGE

Once the NOFO is published, researchers may propose analyses to advance precision medicine, including:

- **Biomarker discovery**
- **Biomarker validation**
- **Novel disease stratification**
- **Multi-omics** for disease risk, diagnosis, and prognosis

All of Us Biospecimens

as of 1/1/2026

14,710,473

Total Aliquots Stored

4,027,144

Total Primary Biospecimens

Blood: DNA, RNA, cfDNA, serum, plasma

Blood: 3,415,973

Urine: 491,151

Saliva: 120,328

Start Using the Researcher Workbench!

Gain Access to the Data for Free

1

Confirm
institutional
access

2

Create an
account & verify
your identity

3

Complete training
& sign our code
of conduct

4

Explore support
resources

5

Use the
Researcher
Workbench

6

Publish
and present
findings

Median of 29 hours for steps 1 to 5

No IRB approval needed, no per-project approvals
Receive \$300 in credits to jumpstart analysis



Join the Workbench



Data passport allows
unlimited access within your
tier

Create, share, and collaborate
on unlimited workspaces

Learn More About Fitbit Data

See data dictionaries and
data release notes with
information on Fitbit device
type data



Questions? Contact support@researchallofus.org

Partnership Opportunities

1. Device Distribution across a Longitudinal Cohort

Provide devices to *All of Us* participants increase data collection and enable health-outcome-linked analysis of device data



2. Offer *All of Us* Enrollment to Device Users

Invite your existing users to join *All of Us* via in-app enrollment or emails, to link their activity to their longitudinal clinical, genomic, and other data



3. Partnered Research Studies

Develop focused studies to study research questions around the scientific impact of device features and health outcomes



4. Data Challenges

Co-host challenges for researchers to use *All of Us* data to develop and validate new wearable-derived metrics and prediction models tied to health outcomes



All of Us Science Day on April 15, 2026

Wednesday, April 15, 2026, is *All of Us* Science Day

Science Day celebrates *All of Us* research discoveries that are leading to real-world health impact for all communities. The event is free, virtual, and open to everyone!

To sign up for Science Day alerts,
scan the QR code or go to
bit.ly/SD26Alerts



Science Day



DRAFT

All of Us Science Day 2026 Agenda
April 15, 2026, 11:00am - 5:30pm EST

Opening Session*	11:00 AM- 11:45 AM (EST)
State of the Science Featured Presentation	
Research Panel 1 & 2 (concurrent)	11:45 AM- 12:30 PM (EST)
Break	12:30 PM - 1:00 PM (EST)
Poster Hall	1:00 PM - 1:45 PM (EST)
Poster Lightning Talks	
Keynote Session*	1:45 PM- 2:15 PM (EST)
Research Panel 3 & 4 (concurrent)	2:15 PM - 3:00 PM (EST)
Workshops (concurrent)	3:00 PM - 3:45 PM (EST)
Research Panel 5 & 6 (concurrent)	3:45 PM - 4:30 PM (EST)
Closing Session*	4:30 PM - 5:30 PM (EST)
Participant Engagement Form	

* Plenary Session



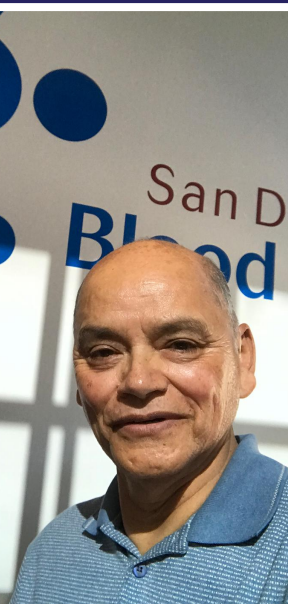
Featured Presentation
Naomi Allen, UK Biobank



Keynote
Pradeep Natarajan,
Massachusetts General
Hospital



ResearchAllofUs.org



AllofUs.NIH.gov

JoinAllofUs.org

Thank you to our 874,000+ participants!